

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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ABSTRACT

The evaluation of generative models remains a defining challenge: because standard automatic metrics demonstrably fail to track human preferences, human judgment endures as the decisive arbiter of comparative quality. That judgment, however, is at once costly to elicit and intrinsically noisy, since annotators differ systematically in expertise, attention, and diligence, and this heterogeneity propagates directly into the recovered rankings. Eliciting relative rather than absolute judgments through pairwise comparison attenuates part of this variability, yet reconstructing a coherent global ordering from a sparse and noisy set of comparisons demands a principled aggregation model. Bradley–Terry estimators are the standard instrument for this task, but existing variants confront an unresolved trade-off: they either presume that every annotator is equally reliable or, when they do model reliability, rely on gradient-based optimization that carries no convergence guarantee, thereby compromising both robustness and interpretability. We resolve this trade-off with BBQ, a Bayesian Bradley–Terry model that jointly and explicitly estimates item quality together with per-annotator reliability, automatically down-weighting or discarding unreliable participants, and that admits closed-form Expectation–Maximization updates with a guaranteed monotonic increase of the observed-data likelihood. Across eight human-preference datasets spanning large-language-model and image-compression evaluation, BBQ delivers the most stable and accurate rankings under noisy and crowdsourced conditions, produces well-calibrated uncertainty estimates, converges rapidly and reliably, and yields interpretable rater-quality parameters that recover precisely which annotators can be trusted. The framework thereby establishes a practical and general foundation for human evaluation of generative models—one that is simultaneously more reliable, more interpretable, and more cost-effective than the prevailing alternatives.

1 INTRODUCTION

The evaluation of generative models is a central and still-unresolved problem, and nowhere more acutely than for large language models (LLMs) and image generators, where standard automatic metrics systematically fail to reflect human preferences. Surrogate metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, afford only limited insight into perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Human evaluation therefore remains the indispensable standard for establishing meaningful rankings among models—which is precisely why contemporary leaderboards and comparative studies of large language models and Learned Image Compression (LIC) methods now rest squarely on human-preference data.

Human evaluation, however, is both expensive and time-consuming, which elevates the design of elicitation protocols that remain efficient while minimizing subjectivity and noise to a first-order concern. Pairwise comparison—in which a participant selects between two items rather than assigning an absolute score—is a particularly effective and practical form of such elicitation (Zerman et al., 2018; Wang et al., 2023), precisely because it imposes a lighter cognitive load and elicits more consistent judgments. Exhaustive comparison is nonetheless infeasible: the number of item pairs grows quadratically in the number of items, so any realistic study observes only a sparse subset of the possible comparisons. In this sparse regime, elementary statistics such as raw win rates are inadequate for recovering overall quality scores, because a win rate irreducibly confounds an

item’s intrinsic quality with the quality of the particular opponents against which it happened to be compared.

Producing a global ranking suitable for a leaderboard therefore hinges on aggregating pairwise comparisons in a principled way—the requirement that motivates the search for robust and efficient aggregation methods. The Bradley–Terry (BT) model (Bradley & Terry, 1952) addresses this problem by iteratively updating item scores from comparison outcomes, and it has become the de facto standard for aggregating pairwise judgments across domains.

Despite its wide adoption, the standard Bradley–Terry model is subject to two limitations that prove decisive in practice. The first is algorithmic. BT parameters can be estimated by iterative scaling (Ford Jr, 1957), gradient descent, or the minorization–maximization (MM) algorithm (Hunter, 2004); the MM algorithm provably converges to the global maximum for the basic BT model, whereas the gradient-based methods closely tied to the Elo rating system, though widely used, offer no comparable guarantee (Hunter, 2004). This guarantee is, moreover, fragile: once the model is extended to accommodate factors such as home-field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may converge only to a local optimum (Hunter, 2004). The second limitation is statistical. Human evaluation is inherently noisy—participants may rush, guess, or lose focus—which yields markedly uneven reliability across raters; when this heterogeneity is ignored, a handful of unreliable raters can distort item scores and destabilize the entire ranking.

We resolve both limitations simultaneously with a Bayesian Bradley–Terry variant that jointly models item quality and rater reliability. To the best of our knowledge, no prior work unifies the BT framework with a Bayesian formulation that explicitly models rater quality while admitting closed-form EM updates: previous approaches to noisy raters depend on gradient-based optimization without convergence guarantees, whereas our formulation delivers stable, guaranteed convergence together with interpretable parameters. Casting the estimator in a Bayesian framework at once captures epistemic uncertainty and supplies principled regularization. To account explicitly for variability in participant behavior, we introduce rater-specific quality parameters that quantify how consistent and trustworthy each participant is, and that allow the model to modulate the influence of individual comparisons in proportion to their estimated reliability. From this latent-variable formulation we derive an Expectation–Maximization (EM) algorithm that jointly and efficiently estimates item and rater parameters in closed form with a guaranteed monotonic increase of the likelihood. The construction lets partially reliable raters continue to contribute useful information while systematically suppressing the influence of inattentive or inconsistent participants. Because it models rater quality within a fully Bayesian framework, the method further furnishes uncertainty estimates for item scores, enabling more interpretable rankings and rigorous comparison across studies.

We establish the effectiveness of our approach on a diverse collection of eight human-evaluation datasets and introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset¹ designed to probe rater reliability directly. Relative to standard BT and naive aggregation, our method converges faster, proves markedly more robust to noisy comparisons, and recovers more consistent rankings. The framework is in no way specific to generative-model evaluation: it applies to the broad class of settings in which noisy pairwise comparisons must be transformed into reliable global rankings. In sum, our work advances human evaluation toward being substantially more cost-efficient, interpretable, and reproducible for assessing AI-generated content.

2 RELATED WORK

Ranking items is a recurring requirement in human studies, and the manner in which judgments are elicited materially affects their quality. Presenting participants with a choice between two items, rather than a single item to be scored in isolation, demonstrably increases sensitivity to subtle differences and reduces response variability (Zerman et al., 2018; Wang et al., 2023). For this reason, many machine-learning leaderboards are built on pairwise comparisons: participants indicate a preference between two alternatives, and these preferences are aggregated into an overall ranking. Prominent instances include the Chatbot Arena (Chiang et al., 2024) and the CLIC image-compression challenge (CLIC, 2025), both of which elicit pairwise comparisons and aggregate them

¹The IHQ dataset will be released upon acceptance.

with variants of the Bradley–Terry (BT) model—a testament to the model’s centrality in modern evaluation pipelines.

The Bradley–Terry model (Bradley & Terry, 1952), originally introduced to rank competitors in sports, provides a probabilistic framework for estimating the probability that one item is preferred over another. By converting pairwise comparisons into a coherent ranking, it has become a cornerstone of studies spanning games, consumer preferences, and numerous other applications, and a substantial body of work has extended it to broaden its reach. The Plackett–Luce model, for instance, generalizes BT from pairwise comparisons to full rankings over multiple items (Plackett, 1975; Luce et al., 1959), defining a distribution over permutations as a product of successive BT probabilities (Luce et al., 1959). Further extensions target specific settings, including home-field advantage (Agresti, 2010), ties (Rao & Kupper, 1967), and comparisons between groups rather than individuals (Huang et al., 2006).

Parameter estimation for the basic BT model is typically carried out by maximum likelihood estimation (MLE). Zermelo (1929) introduced a now widely adopted iterative procedure for computing these estimates, and Lange et al. (2000) subsequently showed that this procedure is a special case of the minorization–maximization (MM) family, which iteratively optimizes surrogate functions to reach a local maximum of the likelihood. Hunter (2004) extended MM to generalized BT models and established the conditions under which convergence to the MLE is guaranteed. For the classical BT model the optimization is convex, so convergence to the global optimum is assured (Hunter, 2004); for extensions such as home-field advantage, multiple comparisons, or ties, however, MM algorithms may converge only to a local maximum (Hunter, 2004). This fragility of the convergence guarantee under model extension is precisely the algorithmic weakness that our formulation is designed to eliminate.

Bayesian formulations of the BT model have been developed to incorporate prior knowledge and regularization (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). In particular, Caron & Doucet (2012) showed that MM algorithms admit an interpretation as Expectation–Maximization (EM) procedures, which in turn enables Bayesian inference via Gibbs sampling by yielding tractable complete-data likelihoods and guaranteeing convergence of the resulting Markov Chain Monte Carlo methods. This Bayesian perspective smooths posterior estimates, reduces susceptibility to local maxima, and quantifies uncertainty in the estimated item skills—the very properties on which we build and that we extend to rater reliability.

A critical line of extension relaxes the assumption that all comparisons are equally reliable by modeling annotator quality directly. Because participant reliability varies widely in crowdsourced evaluation, explicitly accounting for it is not merely desirable but essential (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model that jointly estimates item quality and rater reliability through annotator-specific parameters, placing Gaussian priors on both item and rater parameters and introducing scaling factors in the likelihood to modulate each annotator’s influence. Their posterior inference, however, relies on gradient-based optimization, which is difficult to control in high-dimensional spaces on account of potential local optima and, more fundamentally, offers no convergence guarantee whatsoever. Our EM-based method removes precisely this weakness, guaranteeing monotonic improvement of the likelihood. A simplified variant of the same idea—the Elo-based rater model developed by Google Research—is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and related research (Mentzer et al., 2020; Ballé et al., 2025).

To the best of our knowledge, no prior work unifies the Bradley–Terry model with a Bayesian formulation that explicitly models rater quality while also providing closed-form EM updates; this unification is the distinctive contribution of our work. In contrast to Chen et al. (2013), who couple Gaussian priors with gradient-based optimization that lacks convergence guarantees, we exploit conjugate priors and an EM algorithm that guarantees monotonic likelihood improvement. And in contrast to the Elo-based rater model used in CLIC (CLIC, 2025)—a simplified heuristic built on Elo-style updates—our method delivers a principled Bayesian treatment complete with uncertainty estimates and interpretable rater-quality parameters.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

Transforming a set of noisy pairwise comparisons into a reliable global ranking is a fundamental problem in machine learning and statistics, and it is precisely this problem that our model is designed to solve. Consider a set of K items that are repeatedly compared in pairs by a set of R raters, and let D denote the data comprising the observed outcomes of these comparisons. For any two items i and j , Bradley & Terry (1952) proposed the following model:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ is the parameter associated with item $k \in \{1, 2, \dots, K\}$ and represents its latent skill rating. This model furnishes a clear and interpretable route from a set of observed wins and losses to item scores, but it rests on the simplifying assumption that every comparison is equally reliable. In human evaluation this assumption is routinely violated: participants differ in expertise, attention, and diligence, so their judgments are unequally reliable and, in the limiting case, mutually inconsistent.

To model this heterogeneity in reliability directly, we introduce a rater-specific quality parameter $q_r \in [0, 1]$ that admits a simple and interpretable generative reading: with probability q_r , rater r renders an informed judgment governed by the Bradley–Terry model, while with the complementary probability $1 - q_r$, the rater guesses uniformly at random, as if flipping a fair coin between the two items. This construction yields the following mixture model for the probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The log-likelihood of the data is given by:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ denotes the total number of comparisons of items i and j performed by rater r , and $w_{r,ij}$ the number of those comparisons in which rater r ranks item i above item j . This likelihood admits no closed-form maximizer, so the optimal parameters λ and q cannot be recovered analytically. A generic iterative optimizer such as gradient descent could in principle be applied, but it offers no convergence guarantee and requires careful learning-rate tuning. We therefore adopt the Expectation–Maximization (EM) algorithm, which dispenses with learning-rate tuning entirely and guarantees a monotonic increase of the observed-data likelihood together with convergence to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we interpret the Bradley-Terry model within a Thurstonian framework (Diaconis, 1988). Under this view, a comparison between items i and j is modeled as a race in which each item is assigned a random arrival time, Y_i and Y_j respectively, drawn from independent exponential distributions parameterized by the item skills:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time is declared the winner—a rule that recovers the standard Bradley-Terry probability exactly:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

To render the complete-data likelihood tractable for the EM algorithm, we augment the model with two latent variables. First, we introduce the Bernoulli indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which records whether the c -th comparison of items i and j by rater r was governed by the Bradley-Terry model rather than by a random guess. Using these indicators, we define the latent variable $Z_{r,ij}$ as the sum, taken over the $n_{r,ij}$ comparisons by rater r , of the minimal arrival times of those comparisons governed by the Bradley-Terry model:

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the variable $Z_{r,ij}$ follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma-distributed latent-variable representation is precisely what renders the EM updates tractable in closed form while still accounting for rater-specific quality; it is the structural device on which our closed-form guarantee ultimately rests.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm computes maximum a posteriori (MAP) estimates of the model parameters λ and q by treating the rater-quality indicators and the unobserved arrival times of the Thurstonian interpretation as latent variables, and it is guaranteed to converge to a stationary point of the posterior distribution.

We first specify conjugate prior distributions over the parameters. The item skills λ_k receive a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, which is conjugate to the exponential arrival-time distribution, while each rater-quality parameter q_r receives a Beta prior, $q_r \sim B(\alpha, \beta)$; it is precisely this conjugacy that yields the closed-form updates derived below.

At its core, the algorithm iteratively maximizes the expected complete-data log-posterior conditioned on the current parameter estimates (λ^*, q^*) , alternating between the following two steps until convergence:

E-step: Expectation In the E-step we form the expected complete-data log-posterior, denoted Q : the expectation of the log-posterior over all observed and latent variables, taken with respect to the latent variables conditioned on the observed data and the current parameter estimates. Formally, it is defined as:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood ℓ_c decomposes into three additive components:

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization The M-step updates the parameters by maximizing the Q function. Substituting the expected sufficient statistics computed in the E-step converts the original, intractable optimization into a set of simple closed-form updates. The quantity common to both updates is the posterior probability that a given comparison from rater r was governed by the Bradley-Terry model rather than by chance. This quantity, denoted γ , is exactly the weight assigned to a rater's judgment in proportion to how closely it aligns with the model's current predictions, and is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability that item i beats item j . Thus γ quantifies, under the current parameter estimates, the model's confidence that a comparison was informative rather than random, with larger values marking more trustworthy comparisons.

The update for a rater’s quality q_r takes the form of a regularized weighted average. Its numerator accumulates the rater’s “effective number of wins”—each win weighted by the posterior probability γ that it constituted a meaningful, non-random judgment—and augments this sum with the Beta-prior hyperparameters, which act as regularizers. The denominator normalizes by the total number of comparisons together with the prior parameters. Consequently, the update raises a rater’s quality score precisely when their judgments consistently align with the model’s predictions and lowers it otherwise:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The update for an item’s skill λ_i is a ratio that balances two competing quantities. The numerator aggregates the “effective wins” of item i across all raters, each again weighted by γ , and thereby summarizes the total positive evidence in favor of item i . The denominator accounts for all comparisons in which item i participated and serves as the normalizer. Both terms are regularized by the Gamma-prior hyperparameters, yielding the update:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

The complete derivation of the Expectation-Maximization algorithm, including the E-step expectations and the M-step maximization, is provided in Appendix B.

4 EXPERIMENTAL RESULTS

To evaluate our **Bayesian Bradley-Terry** model with rater **Quality (BBQ)**, we conduct a systematic comparison against the two most representative baselines, each isolating one axis of prior practice: Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012), which models item quality but not rater reliability, and a gradient-descent BT model that does model rater quality (Crowd-BT) (Chen et al., 2013). For Crowd-BT we adopt the Google Research implementation (Google, 2025)—the very code used by both CLIC (2025) and Ballé et al. (2025)—thereby ensuring a faithful and maximally competitive baseline. All hyperparameter settings are reported in Appendix C.

Our evaluation spans eight datasets that jointly cover the two dominant modalities of generative-model evaluation. For language, we use the human-preference benchmarks HUMAINE (Team, 2025) and MT-Bench (Zheng et al., 2023); for image compression, WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), and ConHa (Aczel & Wattenhofer, 2024); and to probe rater reliability directly, we introduce a newly collected inhomogeneous rater quality (IHQ) dataset derived from the CLIC2024 image test set (CLIC, 2025). To isolate the effect of rater quality, we partition IHQ into a screened and an unscreened subset, which together allow us to disentangle how low-quality, high-quality, and mixed comparisons bear on each model. The construction of the IHQ dataset is detailed in Appendix E, and a summary of all datasets—comparisons, raters, and items—appears in Appendix D and Table 2. Because these datasets vary widely in scale, rater count, and per-rater comparison counts, they constitute a broad and deliberately demanding testbed that exercises model behavior across both small, high-quality regimes and large, noisy crowdsourced ones.

Because ground-truth rankings are unavailable for real datasets, we establish reference orderings in two complementary ways. On simulated data, where the true ordering is known by construction, all methods recover it as the number of samples grows, confirming the correctness of every estimator. For the crowdsourced datasets, we adopt the ranking obtained on the full dataset as the reference and verify that it constitutes a sound approximation by inspecting the top-ranked item: on every dataset except IHQ-unscreened, all three models agree on the same top item. The single revealing exception is IHQ-unscreened, where Crowd-BT and Bayes-BT both fail to identify the reference image as the highest-quality item while BBQ recovers it correctly—an early and direct indication of BBQ’s robustness to unreliable raters.

A reliable aggregation method should reproduce the same ordering when a study is repeated. We quantify this stability through bootstrapping, which estimates the variability of the recovered rank-

ings; the procedure is detailed in Appendix F. One methodological caveat warrants explicit statement: the WD and HiFiC studies actively selected comparison pairs—HiFiC via a binary-search strategy and WD via maximum information gain—so the bootstrapping results on these two datasets should be interpreted with commensurate caution.

We report two complementary metrics. Top-1 agreement is the fraction of bootstrapped samples that identify the same best item as the full dataset, and is the metric of interest whenever the identity of the single best model is what matters. Kendall’s Tau (τ) measures the ordinal correlation between rankings (Kendall, 1938), with higher values indicating greater stability, and thereby captures the quality of the ordering as a whole. The two metrics are complementary: the former isolates top-of-ranking accuracy, while the latter assesses the fidelity of the complete ordering.

4.1 PERFORMANCE ACROSS DATASETS

We compare the three models across all eight datasets along two complementary axes—ranking accuracy and ranking stability—summarized by Top-1 agreement and Kendall’s Tau in Table 1. As a general trend, datasets that supply more comparisons per model—MT-Bench, WD, and HiFiC—are easier to rank. HUMAINE constitutes an instructive departure from this trend, underscoring that ranking quality is governed not by the sheer number of comparisons alone but also by factors such as rater consistency and diversity.

Across these datasets, BBQ delivers the strongest overall stability and robustness. It identifies the top-performing item at least as reliably as either baseline on every one of the eight datasets, attaining the best or shared-best Top-1 agreement in all eight cases, and it recovers the overall ranking most accurately (highest Kendall’s τ) on five of the eight datasets, placing a close second on the remaining three. This pattern—never inferior to the baselines on Top-1 agreement, and superior on the majority of datasets for full-ranking accuracy—establishes BBQ as the single most dependable aggregator across the entire testbed.

Crucially, the three datasets on which BBQ ranks second in Kendall’s Tau are precisely those with high-quality, homogeneous rater pools—exactly the regime in which modeling rater reliability is expected to help least, so that a near-tie there is entirely consistent with our account rather than a counterexample to it. On MT-Bench every model perfectly recovers the top item. The WD dataset of Ballé et al. (2025) comprises only five raters—most plausibly the paper’s authors—and thus reflects exceptionally careful evaluation, whereas in IHQ-screened the raters were explicitly filtered for quality. When rater quality is uniformly high, little heterogeneity remains for BBQ to exploit, so its advantage over simpler models narrows to a negligible margin, and the second-place gaps in these cases are correspondingly small. BBQ therefore forfeits essentially nothing where reliability modeling cannot help, and gains substantially wherever it can.

The IHQ experiments, spanning both screened and unscreened subsets, expose the pattern that most sharply separates the methods. When low-quality raters are present, as in IHQ-all and IHQ-unscreened, BBQ decisively outperforms both Crowd-BT and Bayes-BT, reaching 99.32% and 61.92% Top-1 agreement against 85.31%/33.15% for Crowd-BT and 75.07%/24.32% for Bayes-BT—margins of roughly fourteen and twenty-nine percentage points over the stronger baseline in the two noisy settings. All three models attain their highest Top-1 accuracy on the screened subset, confirming the value of rater selection. Notably, although Crowd-BT also models rater quality explicitly, its accuracy degrades sharply on the full dataset, most plausibly because each crowdsourced rater supplies fewer than 40 comparisons, leaving Crowd-BT unable to estimate rater quality reliably and thus exposed to noisy annotations. BBQ, by contrast, retains strong performance without any screening whatsoever, forfeiting only a small margin of Top-1 accuracy and thereby exhibiting genuine, quantifiable robustness to low-quality raters.

Achieving uniformly high rater quality is, moreover, difficult in practice: large-scale crowdsourcing inevitably introduces variability, screening procedures are costly, and subjective factors affect even diligent annotators. It is precisely in this common and consequential regime that BBQ offers a principled means of extracting useful signal from partially reliable raters while systematically suppressing the influence of noisy contributions.

Table 1: Ranking accuracy and stability of the three BT-based aggregation methods (Crowd-BT, Bayes-BT, and our BBQ) across the eight datasets. Top block: Top-1 agreement [%]; bottom block: Kendall’s Tau. Higher is better for both metrics; per-dataset best values are shown in **bold** and second-best underlined. BBQ attains the best or shared-best Top-1 agreement on every dataset, recovers the best overall ranking on five of the eight datasets, and ranks a close second on the remaining three—each of which is characterized by a homogeneous, high-quality rater pool where reliability modeling can contribute little. Its largest margins arise on the noisy IHQ-all and IHQ-unscreened subsets, precisely where modeling rater quality matters most.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	33.15
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	<u>57.30</u>	<u>75.07</u>	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	<u>0.9180</u>	<u>0.9245</u>	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

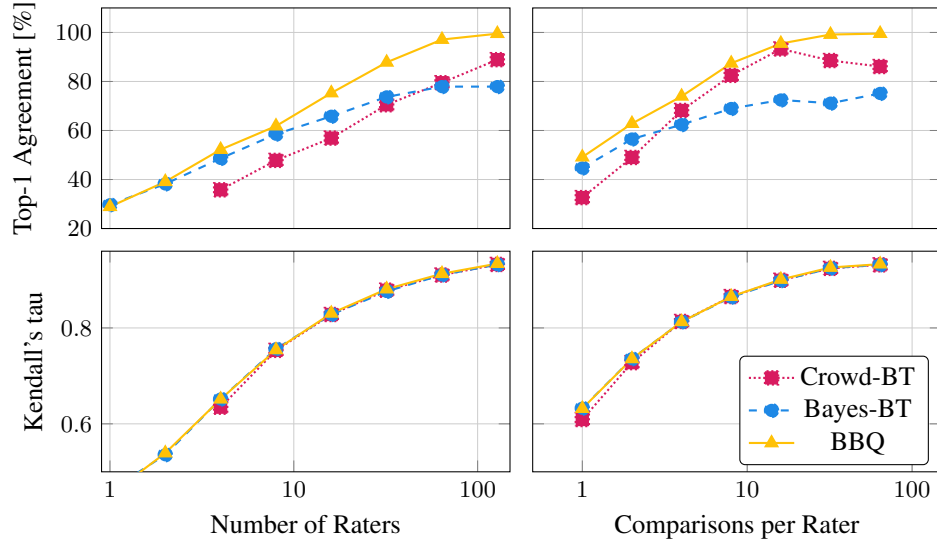


Figure 1: Scaling behavior of the three Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), and BBQ (ours)) on the IHQ-all dataset. **Left:** performance as a function of the number of raters, holding the comparisons per rater at their maximum. **Right:** performance as a function of the number of comparisons per rater, holding the number of raters at its maximum. Both Top-1 agreement (top row) and Kendall’s τ (bottom row) improve markedly as either the number of raters or the number of comparisons per rater grows. Top-1 agreement clearly separates the models, whereas Kendall’s τ remains comparable across them. Crowd-BT fails to converge in the very-few-rater regime, exposing a concrete failure mode of gradient-based estimation and underscoring the reliability of BBQ’s EM updates. Crowd-BT and BBQ behave similarly under sparse data, but BBQ increasingly outperforms Bayes-BT as raters or comparisons accumulate.

4.2 SCALING WITH RATERS AND COMPARISONS

Table 1 shows that datasets with more comparisons per model generally yield better rankings. The total number of comparisons can be increased along two orthogonal axes: by recruiting additional

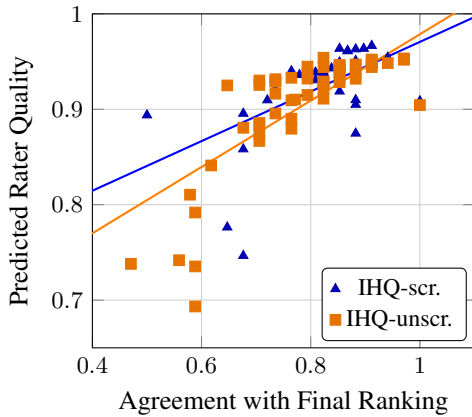


Figure 2: Predicted rater quality (y-axis) against each rater’s agreement with the final ranking (x-axis) for the IHQ datasets, one point per rater; triangles denote the filtered (screened) subset and squares the unfiltered (unscreened) subset. The predicted quality tracks agreement closely in both subsets (Pearson $r = 0.551$ filtered, $r = 0.724$ unfiltered), and several unscreened raters fall below chance-level agreement, confirming that BBQ’s quality parameter faithfully recovers rater trustworthiness.

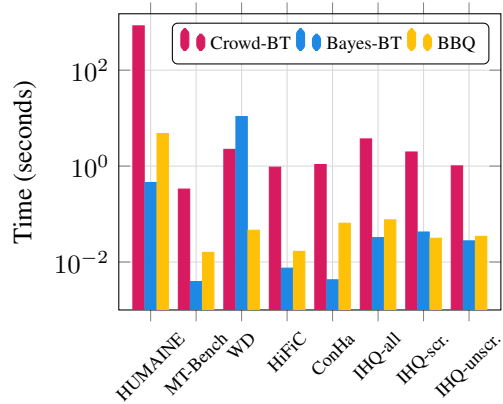


Figure 3: Average computation time in seconds (log scale) to process a single bootstrapped sample, measured for the three models across all eight datasets. Whereas Crowd-BT incurs substantial runtime on several datasets—exceeding 800 seconds on HUMANE—BBQ converges within a few seconds on every dataset, remaining uniformly and consistently fast.

raters, or by soliciting additional comparisons from each rater. Disentangling these axes clarifies where each method’s strengths and weaknesses originate.

Figure 1 isolates the two factors for the three BT variants. The left column varies the number of raters while holding the comparisons per rater at their maximum; the right column varies the comparisons per rater while holding the number of raters at its maximum. The two metrics behave distinctly: Top-1 accuracy separates the models clearly, whereas Kendall’s τ remains comparable across them. Most tellingly, Crowd-BT fails to converge with only one or two raters, exposing a concrete limitation of gradient-based estimation and confirming the practical value of the convergence guarantee that EM affords.

In the sparse-data regime Crowd-BT and BBQ perform comparably, as expected: BBQ requires both multiple raters and multiple comparisons per rater before it can reliably discriminate among rater qualities. Once either quantity grows, however, BBQ increasingly outperforms Bayes-BT. Bayes-BT underperforms overall, yet it can edge out Crowd-BT when the data per rater or the number of raters is very limited, precisely because Crowd-BT cannot estimate rater quality reliably in such sparse settings—a regime in which BBQ nonetheless remains stable.

4.3 RATER QUALITY

A central claim of our approach is that the learned quality parameter is not merely a regularizer but a faithful, interpretable estimate of how trustworthy each rater is. Figure 2 tests this claim directly by relating each rater’s agreement with the final ranking to the quality that BBQ predicts for that rater. The relationship is unambiguously positive: raters who agree more closely with the consensus ranking receive systematically higher predicted quality. The Pearson correlation is $r = 0.551$ across the 50 raters of the filtered dataset and a stronger $r = 0.724$ across the 62 raters of the unfiltered dataset, the higher value reflecting the greater spread of rater reliability once screening is removed.

As anticipated, the unscreened subset contains several raters with both low agreement and low predicted quality, a number of whom fall below chance-level agreement (50%) and thus systematically disagree with the majority. Such adversarial or inattentive raters are exactly the case that motivates explicit rater-quality modeling. BBQ identifies them and assigns them correspondingly low quality

scores, thereby attenuating their influence on the final ranking and neutralizing the noise they would otherwise inject.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average time each method requires to process a single bootstrapped sample across datasets, furnishing a practical measure of feasibility in real-world evaluation pipelines. BBQ converges within a few seconds on every dataset. Crowd-BT is uniformly slower and becomes especially expensive on datasets with many comparisons: on HUMAINE its convergence takes roughly 15 minutes—some two orders of magnitude longer than BBQ. Bayes-BT converges slowly on WD but is otherwise only marginally faster than BBQ. This efficiency is a direct consequence of our closed-form EM updates, which converge rapidly even on large datasets and require no learning-rate tuning whatsoever.

The comparison is, if anything, conservative in BBQ’s favor: Crowd-BT is a heavily optimized C implementation (Kernighan & Ritchie, 1988), whereas BBQ is written in plain NumPy without any specialized optimization, so a compiled or fully vectorized implementation of BBQ could widen the gap still further. BBQ is therefore not only more robust and stable but also eminently practical for large-scale human-preference studies and any application that demands repeated bootstrapping. This combination of accuracy, stability, and speed establishes BBQ as a compelling choice for real-world deployment.

4.5 CONFIDENCE INTERVALS

Beyond point rankings, we assess how faithfully each model quantifies uncertainty over item skills. In a controlled simulation under the null hypothesis H_0 that all items are equally skilled, raters provide pairwise comparisons of items of identical strength, each model estimates skills and constructs uncertainty intervals, and we test for spuriously significant differences (Appendix G). The resulting Type I error rates appear in Figure 6. At the 99% confidence level the expected error rate is 1%. Crowd-BT and BBQ track this target closely, yielding rates slightly below 1%, whereas Bayes-BT is markedly conservative at $\sim 0.1\%$. BBQ therefore delivers well-calibrated uncertainty estimates on par with Crowd-BT, while Bayes-BT systematically underestimates the false-positive rate and is consequently poorly calibrated for significance testing. This calibration property matters in practice: it licenses the use of BBQ’s credible intervals to test whether observed differences between items are statistically significant.

5 LIMITATIONS

While BBQ models rater quality effectively, its advantage over simpler models narrows when raters are uniformly reliable, since a homogeneous rater pool offers little variability to exploit. We emphasize, however, that this boundary is favorable in practice: guaranteeing consistently high-quality raters demands substantial cost and effort and is frequently infeasible at scale, so the regime in which BBQ is most valuable—realistic, noisy crowdsourcing—is also the regime that predominates, whereas carefully curated evaluations remain the exception rather than the rule.

A second limitation concerns the supported feedback modality. BBQ is presently formulated for pairwise comparisons, whereas some human-evaluation studies collect ratings, full rankings, or multi-way inputs; extending the latent-variable formulation to these richer feedback types is a natural direction that would further broaden the framework’s reach.

Methodologically, our stability estimates rest on a finite number of bootstrap resamples. This furnishes a practical and, in our experiments, consistent measure of stability, though larger-scale studies would sharpen the robustness estimates further. The model additionally assumes conditional independence across comparisons and does not model contextual or order effects, which can influence human judgment in practice.

Finally, although BBQ scales favorably, extremely large numbers of items or raters may still prove demanding without a compiled or vectorized implementation—an engineering rather than a methodological constraint, given the closed-form nature of the updates.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that jointly estimates item quality and rater reliability. Its core contribution is an Expectation-Maximization (EM) algorithm that estimates item skills and individual rater quality simultaneously in closed form, automatically down-weighting or removing the influence of unreliable participants. By modeling rater quality explicitly, the method produces more stable and accurate rankings, and its EM formulation guarantees rapid convergence with monotonically improving likelihood—thereby directly resolving the convergence weakness of gradient-based approaches.

These properties translate into consistent empirical gains. Across eight diverse datasets, BBQ attained the strongest overall Top-1 agreement and Kendall’s Tau, and its advantage was largest precisely where it matters most: on noisy and crowdsourced data, where it remained accurate even when individual raters contributed few comparisons. Its edge over simpler models was most pronounced in large-scale settings populated by non-expert raters who vary widely in attentiveness and diligence, confirming that modeling rater quality is decisive whenever evaluation quality is heterogeneous or partially unreliable. The Bayesian formulation additionally furnishes principled, well-calibrated uncertainty estimates for item scores—error bars that can be used to test whether item differences are statistically significant—and the predicted rater quality closely tracks each rater’s agreement with the final ranking, validating BBQ’s ability to identify reliable evaluators.

In sum, BBQ advances human-evaluation methodology by providing a practical, interpretable, and broadly applicable approach to aggregating noisy pairwise comparisons, marking a substantial step toward more cost-effective, interpretable, and reproducible human studies for evaluating AI-generated content.

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A USAGE OF LLMs

In preparing this paper, we employed large language models (LLMs) strictly as supportive tools. ChatGPT, Claude, Gemini, and Grammarly assisted with spellchecking, refining phrasing, and condensing text to improve clarity and readability, while ChatGPT, Claude, and Cursor supported code analysis, code completion, and the generation of visualizations within our development workflow. These tools served purely as auxiliary aids; all core research ideas, experimental design, and interpretation of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider R raters comparing K items, where each rater r carries a quality q_r : with probability q_r the rater follows the Bradley–Terry model, and with probability $1 - q_r$ the rater chooses at random. The derivation that follows instantiates the standard EM procedure (Dempster et al., 1977), incorporating these rater-specific reliabilities into the Bayesian estimation framework and yielding, step by step, the closed-form updates stated in Section 3.

B.1 MODEL DEFINITION

Notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley-Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley-Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley-Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley-Terry probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture with rater quality:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

We need to compute:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

Log-Likelihood of $P(Z \mid A, \lambda)$:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

Log-Likelihood of $P(A \mid q)$:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

Log-Likelihood of $P(D \mid \lambda, A)$:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Complete-Data Log-Likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We introduce conjugate priors: Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and Beta $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

Priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

Final Q -function:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function with respect to the parameters (λ, q) , holding fixed the expectations computed in the E-step, and thereby yields the closed-form updates for each parameter.

Update for q_r The update for each rater quality q_r follows from maximizing Q with respect to q_r , including the contribution of the Beta prior.

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

Update for λ_i The update for each item skill λ_i follows from maximizing Q with respect to λ_i , including the contribution of the Gamma prior.

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we adopt the implementation of Google (2025) together with its default hyperparameters.

The only hyperparameters of the Bayes-BT and BBQ models are the prior-distribution parameters and the convergence threshold. For both models, we declare convergence when no ELO score changes by more than 1 between successive iterations. The ELO score is obtained from the skill parameter λ as:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where we fix the ELO_SCALE_FACTOR to 400.

For the skill parameters in both Bayes-BT and BBQ we place a Gamma prior with shape 5 and rate 0.1, while for BBQ the Beta prior on rater quality uses $\alpha = 10$ and $\beta = 2$.

D DATASETS

We evaluate our models on a diverse set of human-preference datasets spanning both the language and image domains, summarized in Table 2.

For natural language, we use HUMAINE (Team, 2025), a large-scale benchmark with over 100k comparisons, and MT-Bench (Zheng et al., 2023), which supplies model-comparison judgments from crowd workers on multi-turn dialogues.

For image compression, we use three datasets: WD (Ballé et al., 2025), a dense, expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which targets conditional generative models. These datasets differ substantially in scale, rater count, and rater expertise, and thereby constitute a broad testbed for robustness.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, collected in a user study on the CLIC2024 (CLIC, 2025) data via the *Mabyduck* platform. It comprises two-alternative-forced-choice (2AFC) judgments over 28 generative image-compression models. To isolate the impact of rater quality, we provide two subsets: (i) a **screened** subset, in which raters passed pre-screening checks for attention and display quality, and (ii) an **unscreened** subset containing all raters. The screened subset represents high-quality raters, whereas the unscreened subset deliberately reproduces the noisy conditions typical of large-scale human evaluation.

Taken together, this collection allows us to study both large-scale, relatively clean benchmarks and smaller, noisier settings in which modeling rater reliability becomes critical.

Table 2: Summary of the datasets used in our experiments, reporting the number of comparisons, raters, and models together with the average comparisons per rater and per model. The IHQ dataset was collected by us on the Mabyduck platform and includes both 2AFC and 3AFC settings. For IHQ we provide screened and unscreened subsets: the screened subsets retain only raters who passed pre-screening tests for attention and display quality, thereby ensuring higher reliability, whereas the unscreened subsets include all raters.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

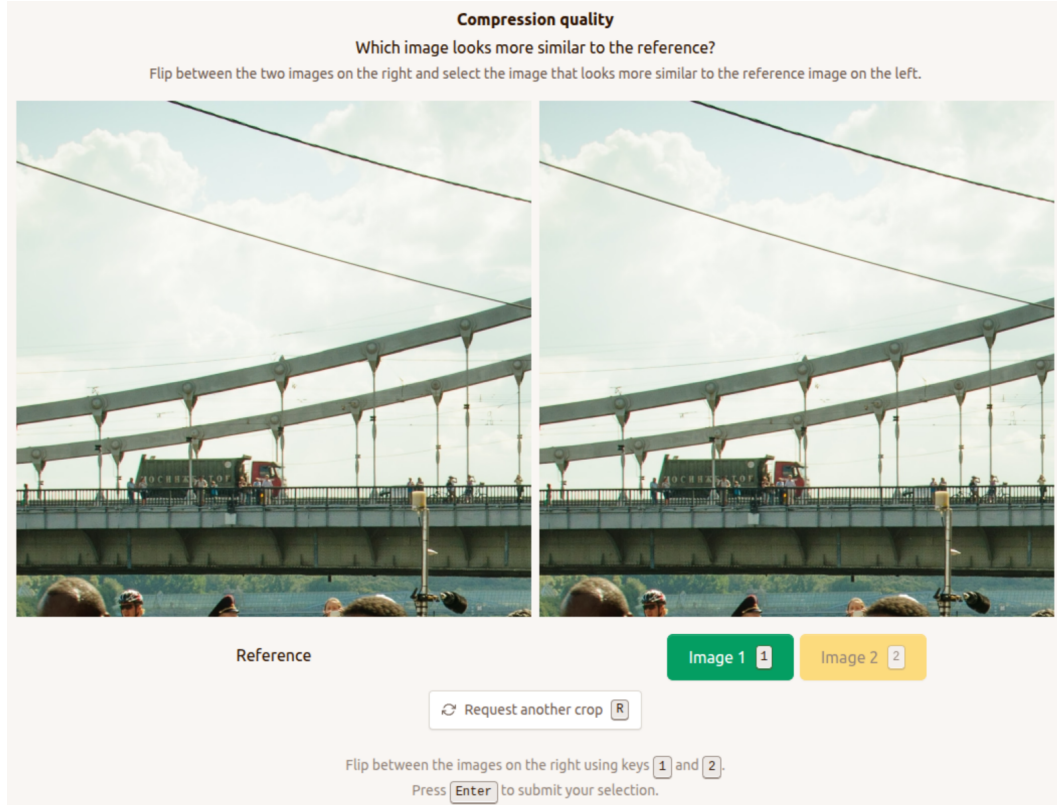


Figure 4: The Mabyduck user-study platform used to collect the pairwise comparisons. The reference image is displayed on the left, while the rater selects between the two compressed candidates shown on the right.

E USER STUDY PLATFORM

All pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset were collected via the Mabyduck platform (Ltd., 2025), shown in Figure 4. For each comparison the platform poses the question “Which image looks more similar to the reference image?”; ties are disallowed, and all pairs were sampled uniformly at random, ensuring an unbiased coverage of the comparison space.

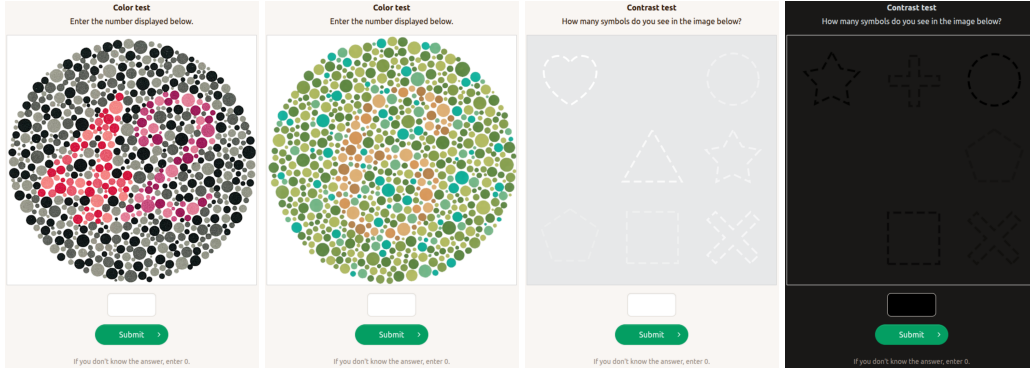


Figure 5: Four representative images from the rater pre-screening process. The first two are color-blindness tests, in which raters must identify the number embedded in each pattern; the last two are shape-detection tests that assess sensitivity to low-contrast objects—a light gray shape on a white background and a dark gray shape on a black background.

To study rater quality directly, we partition the IHQ dataset into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and additionally completed training comparisons in which they selected the higher-quality image, so as to reinforce the evaluation criteria.

Figure 5 presents four representative pre-screening images. The first two are standard color-blindness tests, in which raters must correctly identify the number shown in each pattern, while the final two are shape-detection tests that gauge sensitivity to low-contrast objects—one presenting a light gray shape on a white background and the other a dark gray shape on a black background. Together these tests ensure that only raters with adequate visual acuity contribute to the screened subset, thereby rendering the screened/unscreened distinction a controlled proxy for rater quality.

F BOOTSTRAPPING DETAILS

Because both BBQ and Crowd-BT model rater quality explicitly, we bootstrap over raters rather than over individual comparisons; this preserves each rater’s comparison distribution and thereby yields a fair assessment of stability. For every dataset we draw 10,000 bootstrap resamples of raters, with the sole exception of HUMAINE: as discussed in Section 4.4, a single Crowd-BT bootstrap iteration on HUMAINE takes roughly 15 minutes, so we reduce the count to 1,000 resamples there in order to keep computation tractable.

G UNCERTAINTY ESTIMATION DETAILS

Beyond producing a ranking, quantifying uncertainty is essential for judging whether observed differences between items are statistically significant.

The Bayesian Bradley-Terry (Bayes-BT) and Bayesian Bradley-Terry with Quality (BBQ) models quantify this uncertainty through the posterior distribution over item skills: each skill carries a Gamma prior that is updated by the observed pairwise comparisons, and the resulting posterior credible intervals are mapped to the Elo scale so as to enable direct comparison across items.

The classical Crowd-BT model, being non-Bayesian, instead estimates uncertainty through a frequentist approximation based on a second-order Taylor expansion around the maximum likelihood estimate. The Hessian of the log-likelihood is inverted to obtain the covariance matrix, whose diagonal entries yield the per-item variances. These variances are converted to 99% confidence intervals via:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

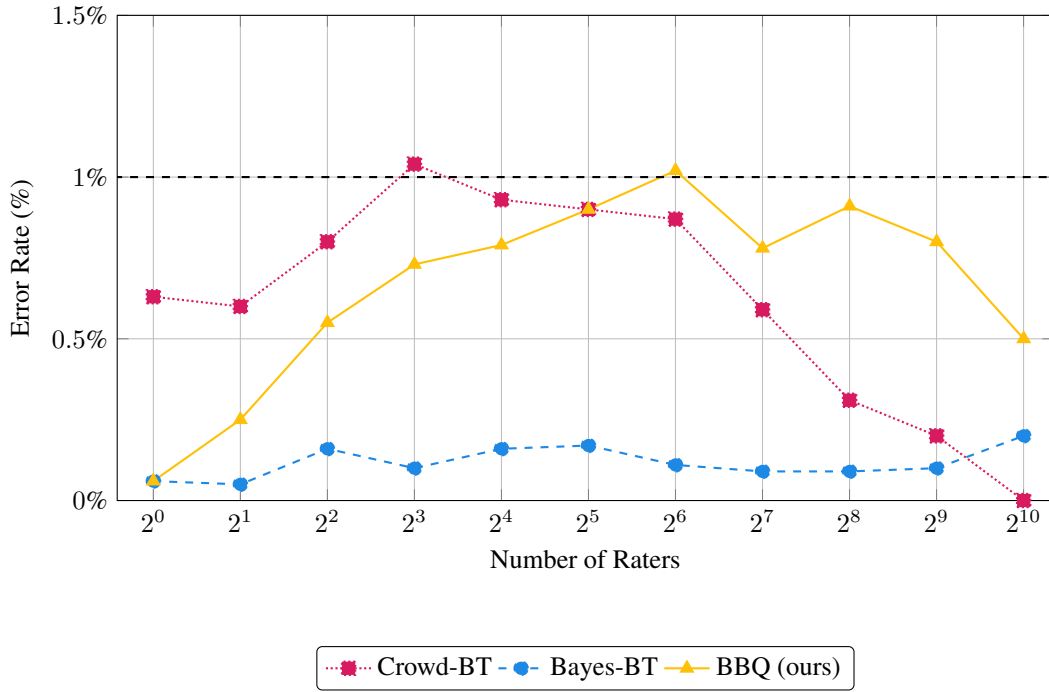


Figure 6: Type I error rate as a function of the number of raters, with the dashed line marking the expected 1% error rate at the 99% confidence level. Crowd-BT and BBQ track the target closely, confirming well-calibrated uncertainty estimates, whereas Bayes-BT is markedly conservative and thus poorly calibrated for significance testing.

where the constants scale the standard deviation to the 99% confidence level. Narrower likelihood peaks yield smaller intervals, reflecting higher certainty, whereas flatter peaks yield larger intervals, reflecting greater uncertainty.

To evaluate how faithfully each model estimates uncertainty, we conduct a controlled simulation. We generate trials comprising two items of equal skill (a 50–50 win probability) and synthesize pairwise-comparison data by coin flips; within each trial we simulate multiple raters and apply the models to estimate Elo scores together with their associated uncertainty.

Formally, the null hypothesis H_0 is that the two items are equally strong. For each trial we compute the 99% confidence (or credible) interval of each item’s skill estimate, and we deem a model to have incorrectly rejected H_0 whenever these intervals fail to overlap—that is, whenever it reports a significant difference where none exists. The metric of interest is therefore the frequency of such incorrect rejections, namely the observed Type I error rate.

We run 10,000 trials with 50 comparisons per rater while varying the number of raters, which allows us to examine how the calibration of each model’s uncertainty estimates scales with the amount of data and to compare the empirical Type I error rate against the theoretical 1% at the 99% confidence level. As Figure 6 shows, Crowd-BT and BBQ maintain Type I error rates close to the expected 1%, confirming that their uncertainty estimates are well-calibrated, whereas Bayes-BT is systematically over-conservative, producing consistently lower error rates and hence credible intervals that are poorly calibrated for significance testing.